Accurate Registration of Multitemporal UAV Images Based on Detection of Major Changes

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***Abstract*—accurate registration of multitemporal images captured by UAV usually involves affine transformation and complicated non-rigid transformation, which makes it very difficult to achieve satisfying results. Pixel-wise correspondence is effective for handling images with complicated non-rigidity. However, objects with changes in the scene deform severely because there should be no pixel-wise correspondence but the algorithm erroneously matches the pixels. In this paper, we propose a coarse-to-fine registration method for multitemporal UAV images. First, a projective model is used to eliminate large scale changes as well as perspective distortion. Then the major changes of different temporal UAV images are most detected, which is used to mask the dense matches in changed areas. Finally, the optical flow field method is used to handle complicated non-rigid changes by matching dense SIFT feature. Experimental results on a challenging set of multitemporal UAV images demonstrate the effectiveness of our approach.**

***Index Terms*—multitemporal image registration, change detection, projective transformation, optical flow field**

I. Introduction

Unmanned Aerial Vehicles (UAVs)-based imaging systems have been rapidly developed among low altitude remote sensing in recent years, which have several advantages over traditional aerial photography, such as higher flexibility and lower cost in collecting image data. However, because UAV systems are usually unstable and work in complex environment, affine transformation and complicated non-rigid changes often happen in the captured images during time series observation of the same scene. In addition, scene changes may occur between the times the images are acquired. Therefore, multitemporal UAV image registration is a challenging task and a critical prerequisite in a wide range of applications including image fusion, change detection, object extraction, etc.

The task of image registration is to find an optimal geometric transformation between corresponding images. Based on the nature of geometric transformation, existing image registration methods can be generally divided into two categories: parametric and non-parametric. The transformation of parametric image registration is governed by a finite set of image features or by expanding the transformation in terms of basis functions [1]. Methods based on local invariant features, such as points [2], lines [3] and edges [4], have the advantage of being robust to typical appearance changes and large scene movements. Basis function based methods based on mutual information [5], cross-correlation [6] and fast Fourier transform [7], have high registration accuracy but low robustness to significant changes in scale and rotation. One of the widely used transformation model is projective transformation, which fits affine transformation and local distortions caused by imaging viewpoint changes but is less effective for handling complicated non-rigidity. Thin plate Spline (TPS) [8] is another commonly used transformation model, which fits affine transformation and elastic distortion based on corresponding feature point set. Its effective for handling significant non-rigidity but registration accuracy relies heavily on the distribution of feature points. Non-parametric image registration originates from modelling the transformation as physical displacement. Optical flow filed is a widely used physical model that every pixel has its own displacement vector, thus it can handle complicated non-rigid changes. Optical flow techniques [9] [10] compute motion field by directly minimizing pixel-to-pixel dissimilarities. The typical objective function of optical flow is built on assumptions including brightness constancy and piecewise smoothness of the pixel displacement field. Hence, it only works well for very similar images. Based on the computational framework of optical flow, Liu et al. [11] proposed scale-invariant feature transform (SIFT) flow by matching SIFT descriptors instead of raw pixels. The SIFT flow algorithm has demonstrated impressive pixel-wise correspondence results for handling complicated non-rigid changes, especially when there is drastic appearance changes due to phenomena such as changes of seasons and variations of imaging conditions (angle, lighting, sensor). However, its not robust to significant changes in scale and rotation because the dense SIFT descriptor is computed at a fixed orientation and scale. Additionally, because there should not be pixel-wise correspondence in changes while the SIFT flow algorithm wrongly match the pixels, the moving objects deform severely, which is not allowed for subsequent application. And it also affects the matches around the changes, which further deteriorates the registration results.

To register multitemporal UAV images accurately and to avoid deformation of objects without correspondence, we propose a coarse-to-fine image registration method. First, images are roughly registered using the projective transformation to handle affine transformation and imaging viewpoint change, which regulates the large variations of scale and rotation and makes it available to detect the main changes of different temporal UAV images. We choose SIFT to match feature points and RANSAC to compute the transformation parameters. Then

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the major changes are detected by object based image analysis. Finally, images are accurately registered using the optical flow field to handle complicated non-rigid changes by masking the matches in changes. Experimental results on a challenging set of multitemporal UAV images show that our method enables accurate registration and disables the deformation of objects.

The rest of this paper is organized as follows. In section II, the proposed method is described in detail. Experimental results are presented in section III. Finally, conclusion is drawn in section IV.

II. METHODOLOGY

Given a pair of different temporal UAV images, our goal is to implement high-accuracy image registration between them. The flowchart of proposed coarse-to-fine registration method is presented in Fig.1. The core steps can be divided into two aspects: coarse registration with projective transformation and fine registration with optical flow field. The coarse registration makes it available to detect major changes and reduce the affection of large scale and rotation changes. The fine registration includes major change detection and finding the optimal optical flow field. These two aspects are described in detail in the following sections.

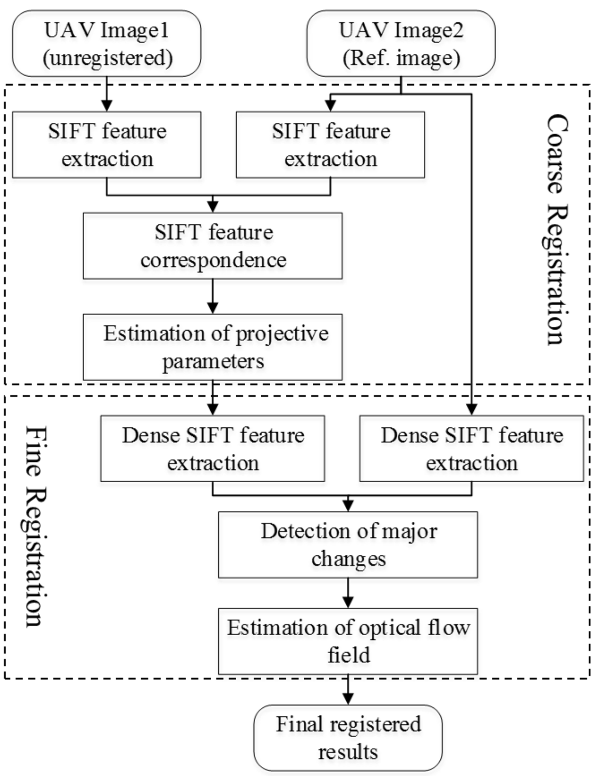


Fig 1. Flowchart of the proposed method.

A. Coarse Registration

We first choose projective transformation to mainly handle global affine transformation and change of perspective between two UAV images. A point $\boldsymbol{p} = (x, y)$ in a plane is represented in homogeneous coordinates by a vector of 3 coordinates, $[x_h, y_h, w_h]^T$, and both coordinates are related by $x = x_h/w_h$ and $y = y_h/w_h$. Transformed points $[x_h, y_h, w_h]^T$ and original points $[i, j, 1]^T$are related by:

$$[x_h, y_h, w_h]^T = H[i, j, 1]^T \tag{1}$$

$$H = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & 1 \end{bmatrix} \tag{2}$$

The matrix H for projective transformation has 8 parameters, which can be solved with 4 pairs of feature points.

We select SIFT algorithm to match feature points for its scale, rotation and illumination invariance. Then we utilize RANSAC algorithm to compute projective transformation parameters. It randomly selects four correspondences to compute parameters and counts the number of correspondence in line with the computed projective transformation, which considers the parameters with the most correspondences as the optimal parameters.

B. Detection of Major Changes

Optical flow field is utilized to handle complicated local non-rigid changes, which is computed by matching dense SIFT descriptor. We denote the dense SIFT descriptor of the reference image and the roughly registered image as $\boldsymbol{s}_1$ and $\boldsymbol{s}_2$. Its robust to variations of imaging conditions. The object movement will affect registration accuracy because nonexistent pixel-wise correspondences in changes are wrongly matched by SIFT flow algorithm, which will affect the optimization of objective function and deform the object with changes. Thus, major changes are first detected. The data term of the objective function of SIFT flow is modified based on the results of change detection to find the optimal optical flow field.

Because of low accuracy of coarse registration, two images are not aligned completely. Thus, we detect the major changes based on superpixel partition [12]. We use the Simple Linear Iteration Clustering (SLIC) segmentation algorithm [13] to segment the roughly registered image into m superpixels $\{S_t^1, S_t^1, \ldots, S_t^m\}$. Then the superpixel segmentation result is applied to the reference image to keep the correspondence of superpixels. SLIC can produce consistent superpixels with similar size and shape, as well as preserve objects boundaries.

After image partition, the description of superpixels is an important task, which directly relates to the measurement of similarity between superpixels. Considering that dense SIFT descriptors will be used in pixel-wise correspondence and the descriptor of pixels in changes differs a lot, we utilize the mean and variance of dense descriptors in a superpixel to represent the structure information of the superpixel. Besides, to increase discriminative power, we combine color feature and structure feature to describe each superpixel. We apply the discriminate color descriptor (DCD) [14] to represent the color information of the superpixel. DCD is learned from natural image dataset in Google, which can automatically maintain photometric invariance to some extent.

Structure descriptor and color descriptor are concatenated to constitute the descriptor of superpixel. The two types of descriptors should be normalized before being concatenated. The similarity of corresponding superpixel is measured by the Euclidean distance. Then the maximum entropy thresholding algorithm [15] is utilized to determine the major changes. We use $c(\boldsymbol{p})$ to indicate whether pixel $\boldsymbol{p}$ belongs to the major changes:

$$c(\boldsymbol{p}) = \begin{cases} 0, & \boldsymbol{p} \in \text{changes} \\ 1, & \text{others} \end{cases} \tag{3}$$

The detected major changes will be used in the following procedure of finding the optimal optical flow field.

C. Finding the Optimal Optical Flow Field

We modify the data term of the objective function of SIFT flow to find the optimal optical flow field based on the results of change detection.

The SIFT flow is inspired by optical flow methods which are able to produce dense, pixel-to-pixel correspondences between two images. It is very suitable to handle the complicated non-rigid changes of multitemporal UAV images. Let $\boldsymbol{\omega}(\boldsymbol{p}) = (u(\boldsymbol{p}), v(\boldsymbol{p}))$ be the flow vector at $\boldsymbol{p}$ and w be the spatial neighborhoods. The energy function for SIFT flow is defined as:

$$\begin{aligned} E(\boldsymbol{\omega}) = &\sum_{\boldsymbol{p}} \boldsymbol{min}(\|\boldsymbol{s}_1(\boldsymbol{p}) - \boldsymbol{s}_2(\boldsymbol{p}+\boldsymbol{\omega}(\boldsymbol{p}))\|_1, t) \\ &+ \sum_{\boldsymbol{p}} \eta(|u(\boldsymbol{p}) + v(\boldsymbol{p})|) \\ &+ \sum_{\boldsymbol{p},\boldsymbol{q}\in w} \boldsymbol{min}(\alpha|u(\boldsymbol{p}) - u(\boldsymbol{q})|, d) \\ &\quad + \boldsymbol{min}(\alpha|v(\boldsymbol{p}) - v(\boldsymbol{q})|, d) \end{aligned} \tag{4}$$

where the three terms are the data term, small displacement term, and smoothness term, respectively. The data term constrains the SIFT descriptors to be matched along with the flow vector $\boldsymbol{\omega}(\boldsymbol{p})$. The small displacement term constrains the flow vectors to be as small as possible when no other information is available. The smoothness term constrains the flow vectors of adjacent pixels to be similar. In this objective function, truncated L1 norms are used in both the data term and the smoothness term to account for matching outliers and flow discontinuities, with t and d as the threshold, respectively.

However, the use of threshold t fails to account for matching outliers effectively. Because the structure features around changes may be similar to the changes. Thus we modify the data term of objective function of SIFT flow based on the results of change detection to disable the pixel-wise matches in major changes:

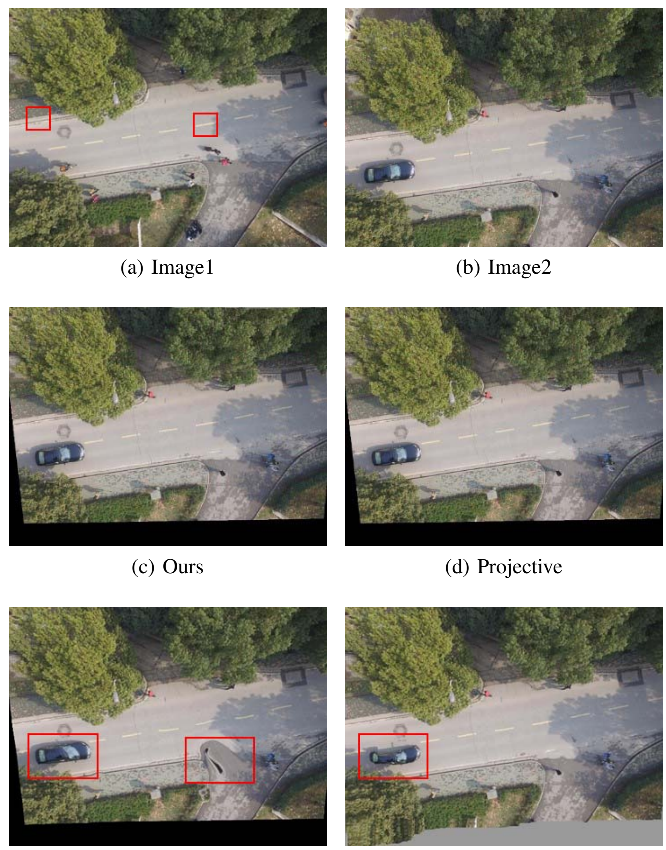
$$\sum_{\boldsymbol{p}} c(\boldsymbol{p})\boldsymbol{min}(\|\boldsymbol{s}_1(\boldsymbol{p}) - \boldsymbol{s}_2(\boldsymbol{p}+\boldsymbol{\omega}(\boldsymbol{p}))\|_1, t) \tag{5}$$

By ignoring the correspondence in changed areas, objects with movement have no deformation and the correspondence in areas around changes also avoid the affection of changes. Simultaneously, because of the existing small displacement term and smoothness term, the displacement vector of pixels in changes will be similar with the surrounding area.

III. EXPERIMENTAL RESULTS

We evaluate the performance of our method on a challenging set of multitemporal UAV images. At the times of imaging, the flight attitude and height of UAV as well as the shooting angle may be different. The imaging scene are most non-planar and changes may happen during time series observation. We compare our method with projective-based [16], TPS-based [8], and SIFT flow [11] methods qualitatively and quantitatively. We compute the projective transformation and TPS transformation with the same correspondences matched by SIFT algorithm, in which outliers are removed by RANSAC algorithm. And we tune the parameters of SIFT flow algorithm to find the optimal settings.

Here we demonstrate the registration results on two typical examples, as shown in the first two images in Fig. 2 and Fig. 3. For the first pair of images, texture of imaging scene mainly concentrates on leaves and the texture on pavement is very weak. The changes are relatively few, centering on

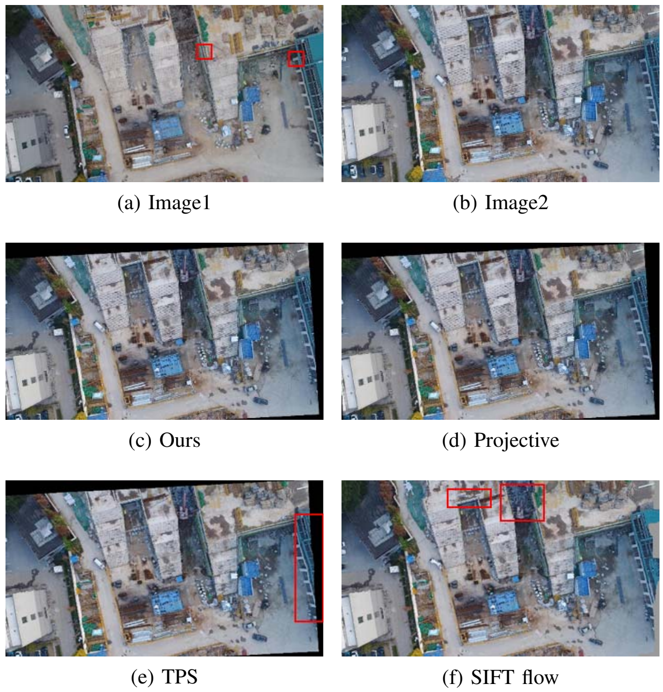


(a) Image1 (b) Image2

(c) Ours (d) Projective

(e) TPS (f) SIFT flow

Fig 2. Qualitative comparison of registration results on multitemporal UAV image pair 1. The goal is to align the test image (b) onto the reference image (a); (c)-(f) show the results of our coarse-to-fine method, projective-based method, TPS-based method and SIFT flow respectively.

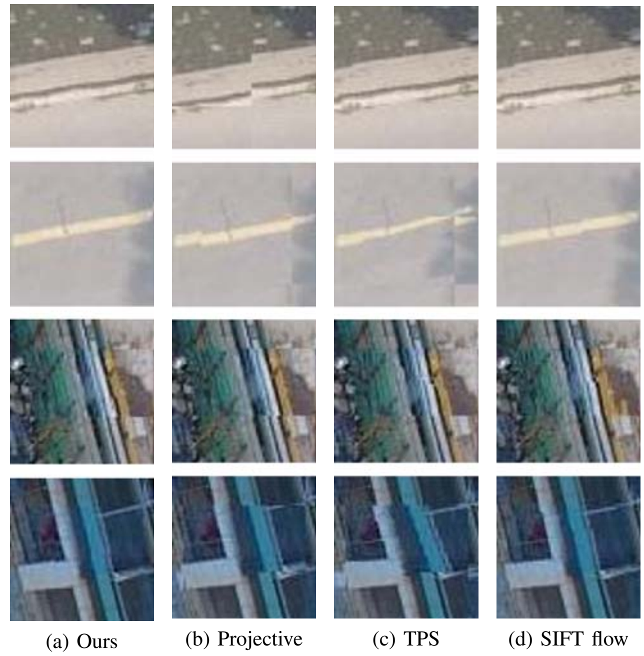


(a) Image1 (b) Image2

(c) Ours (d) Projective

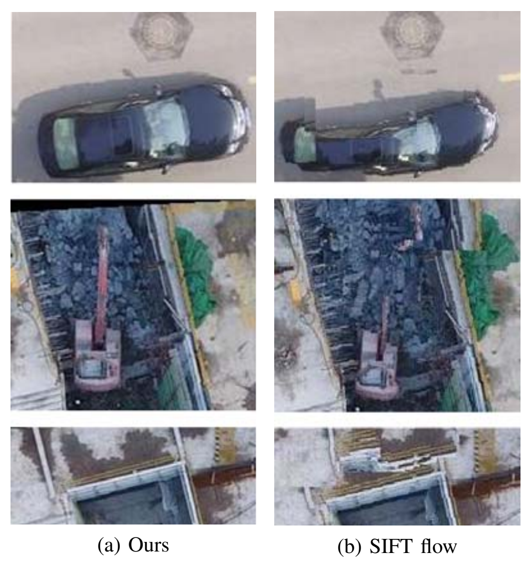
(e) TPS (f) SIFT flow

Fig 3. Qualitative comparison of registration results on multitemporal UAV image pair 2. The goal is to align the test image (b) onto the reference image (a); (c)-(f) show the results of our coarse-to-fine method, projective-based method, TPS-based method and SIFT flow respectively.



(a) Ours (b) Projective (c) TPS (d) SIFT flow

Fig 4. The details of the selected regions marked by the rectangles in Fig. 2(a) and Fig. 3(a). From left to right are the results of ours, projective-based, TPS-based and SIFT flow, respectively.



(a) Ours (b) SIFT flow

Fig 5. The details of the selected regions marked by the rectangles in Fig. 2 (f) and Fig. 3 (f) . From left to right are the results of ours and SIFT flow, respectively.

moving vehicles and pedestrians. For the second pair, the color information of images differs greatly because of the phenomena change and there are plenty of changes because the buildings are under construction. The registered results are also presented in Fig. 2 and Fig. 3. In each figure, (c), (d), (e), and (f) are the registration results of our method, projective-based method, TPS-based method, and SIFT flow, respectively.

From the results, we can find that some areas in the registered images of the TPS-based method and the SIFT flow method are seriously out of shape. The out-of-shape areas of the TPS-based method concentrate on regions where matched feature points are very few, such as the selected regions marked with red rectangle in Fig. 2(e) and Fig. 3(e). The texture is very weak or highly repeated, thus it is difficult to extract or accurately match feature points. And the out-of-shape areas of the SIFT flow method scatter over the regions with changes where pixels without correspondence are wrongly matched. We also select several regions in Fig. 2(f) and Fig. 3(f) to present.

To demonstrate the alignment degree of the four methods intuitively, we use checkboard displays of the registered results in Fig. 4 by zooming into the details. The front two rows are the results of the first image pair and the rest two rows are the results of the second image pair. The regions we choose to display are marked with the rectangle in Fig. 2(a) and Fig. 3(a) respectively. We can clearly observe that images are not aligned based on projective transformation. Taking the first image pair as example, the pavement is not aligned because the feature points are mainly concentrated on leaves with few on pavements and the leaves are not in a plane with the pavement. The corresponding regions are also partially aligned based on

TPS transformation. We can find the accuracy of registration depends heavily on the distribution of feature points. Areas with densely distributed feature points are registered accurately but areas with few feature points are registered poorly. These two methods operating on a set of feature points rather than on the whole pixels have problems in case of complex non-rigid deformations. SIFT flow performs well on the four selected regions. However, in the regions with changes, the performance of SIFT flow degrades severely. In Fig. 5, we show the details of the warpped images in regions marked in Fig. 2(f) and Fig. 3(f). The moving objects are badly twisted in the result of SIFT flow. Additionally, there are many error matches in surrounding areas with no-changes. Our proposed method performs well in the scene alignment and shape maintenance of objects with changes.

We also quantitatively compare the accuracies of the four methods by root mean square error (RMSE), as shown in TABLE I. Clearly, our method has the lowest RMSE and achieves subpixel accuracy. The Projective-based and TPS-based methods fail to accurately register images with complicated non-rigid distortions because they are based on a finite number of feature points. The accuracy of SIFT flow is lower than ours due to the wrong matches in change areas and the poor ability of handling large variants of rotation and scale.

TABLE I. Quantitative comparison of registration results on two pairs of multitemporal UAV images with RMSE

	Ours	**Projective-based**	**TPS-based**	**SIFT flow**
Image pair 1	0.4395	2.9002	1.6170	0.8738
Image pair 2	0.5722	2.4093	4.6936	1.0210

Fig 6. Registration results of UAV image sequences (8 images) by our proposed method.

Fig 7. Registration results of UAV image sequences (8 images) by our proposed method.

We further validate our method on time series of UAV images. The results are shown in Fig. 6 and Fig. 7. We choose the first image as reference image and the rest of images are aligned onto the reference image. Our method performs well in the scene alignment and shape maintenance of objects with changes.

IV. CONCLUSION

In this paper, a coarse-to-fine image registration method has been proposed to align multitemporal UAV images. It follows a two-step strategy: the first step eliminates large affine transformation and change of perspective using a projective model; the second step approximates the complicated non-rigid deformation with optical flow field. The estimation of flow field is improved by a modified data term of SIFT flow based on masking the pixel-wise matches in major changes, which are detected with the coarsely registration result in the first step. The qualitative and quantitative experimental results on challenging multitemporal UAV images demonstrate that our method achieves higher registration accuracy and shows better ability of shape maintenance than the competitors.

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